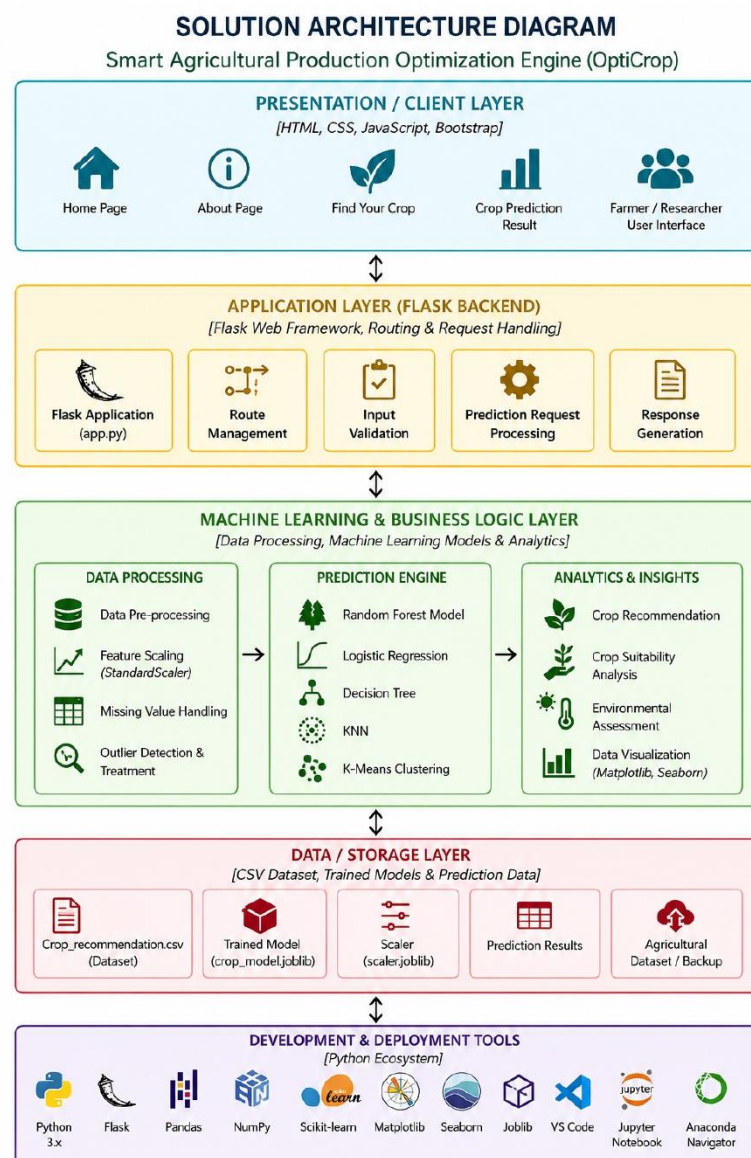


Solution Architecture

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where farmers enter soil nutrient and environmental parameters and view the recommended crop. Handles user interactions and displays prediction results.	HTML5, CSS3, JavaScript
API Gateway	Receives user requests from the frontend, validates the input, forwards the data to the machine learning prediction module, and returns the prediction response.	Flask (Python), REST API, JSON
Microservices	Implements the core business logic, including data preprocessing, feature scaling, loading the trained machine learning model, and generating crop recommendations based on the input parameters.	Python, Scikit-learn, Joblib, NumPy, Pandas
Database	Stores the agricultural dataset used for model training and the serialized machine learning model for efficient prediction without retraining.	Crop_recommendation.csv, Joblib (.joblib/.pkl) model files